



DEPARTMENT OF MATHEMATICS

Sub Code: MAC21		Sub: Numerical Techniques and Differential Equations		Section: A to L
Semester: II		Term: 01/03/2024 to 22/06/2024		Marks: 30
Date: 18/04/2024		Time: 9:30 to 10:30 am		Test: I

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Q. No.	Questions	Blooms Level	CO's	Mark s
1.	(a) Write the Taylor series expansion of a function $f(x,y)$ of two variables up to second degree terms.	L1	CO1	2
	(b) Find the orthogonal trajectories of the family of circles $r = a$.	L2	CO2	3
	(c) Find a real root (other than 0) of the equation $x^3 - 2x + \sin x = 0$, correct to three decimal places using Newton-Raphson method.	L3	CO1	5
	(d) Using Euler's modified method, solve the following initial value problem: $\frac{dy}{dx} = \log_{10}(x+y)$; $y(0)=1$ at $x=0.2$, $h=0.1$ (perform two iteration at each stage).	L4	CO2	5
2.	(a) Write the differential equation of closed circuit involving L, C, and R in series with applied e.m.f.	L1	CO2	2
	(b) Expand $e^{2x} \cos 3x$ as Maclaurin's series up to third degree terms.	L2	CO1	3
	(c) Solve the following initial value problem using Runge-Kutta method of fourth order: $\frac{dy}{dx} = xy - x^2$; $y(1)=0.5$ at $x=1.1$ taking step length of 0.1.	L3	CO2	5
	(d) A wire of length 24 meter is cut into two parts which are bent in the form of a square and circle respectively. Find the least value of the sum of the areas so found.	L4	CO1	5
3.	(a) Define stationary and saddle point.	L1	CO1	2
	(b) The temperature of a cup of coffee is 98°C , when freshly poured at the room temperature being 30°C . In five minutes it was cooled to 66°C . How long a period must elapse, before the temperature of coffee becomes half of its initial temperature?	L2	CO2	3
	(c) Solve the following system of non-linear equations using Newton-Raphson method (Carry out one iteration) $x^2 + y^2 = x$, $x^2 - y^2 = y$, given that $x_0 = 0.8$ and $y_0 = 0.4$.	L3	CO1	5
	(d) Find the orthogonal trajectories of the family of curves $x+y = ce^y$, where c is the parameter.	L5	CO2	5